```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('ecommerce_raw_dataset.csv')

print("Dataset loaded successfully!")
print("Shape:", df.shape)

df.head()
```

```
Dataset loaded successfully!
Shape: (51, 12)
```

```
   Order_ID  Order_Date    Customer_Name               Product
Category  \
0  ORD-1001  2025-01-01    nEHa sharma         Wireless Mouse
Electronics
1  ORD-1002  2025-01-04      Neha Sharma   bluetooth speaker
Clothing
2  ORD-1003  2025-01-07       Ravi Kumar            Rice 5 Kg
eLeCtronics
3  ORD-1004  2025-01-10       Isha Reddy       Ballpoint Pens
Stationery
4  ORD-1005  2025-01-13      Karan Patel            Desk Lamp  Home &
Kitchen

  Quantity Unit_Price Discount_Pct Payment_Method         City
Order_Status  \
0        1     799.00            0            UPI    Hyderabad
Delivered
1        2     569.05            5    Credit Card      Chennai
Processing
2        1     418.95           10     Debit Card    Bengaluru
Returned
3        3      99.00           15   credit CARD        Mumbai
Shipped
4        2    1079.10           20    Net Banking   HYDERABAD
Cancelled

  Shipping_Days
0             1
1             2
2             3
3             4
4             5
```

```
df.head(5)
```

```
   Order_ID  Order_Date    Customer_Name               Product
Category  \
```

```
0  ORD-1001  2025-01-01    nEHa sharma         Wireless Mouse
Electronics
1  ORD-1002  2025-01-04      Neha Sharma   bluetooth speaker
Clothing
2  ORD-1003  2025-01-07       Ravi Kumar            Rice 5 Kg
eLeCtronics
3  ORD-1004  2025-01-10       Isha Reddy       Ballpoint Pens
Stationery
4  ORD-1005  2025-01-13      Karan Patel            Desk Lamp  Home &
Kitchen

  Quantity Unit_Price Discount_Pct Payment_Method         City
Order_Status  \
0        1     799.00            0            UPI    Hyderabad
Delivered
1        2     569.05            5    Credit Card      Chennai
Processing
2        1     418.95           10     Debit Card    Bengaluru
Returned
3        3      99.00           15   credit CARD        Mumbai
Shipped
4        2    1079.10           20    Net Banking   HYDERABAD
Cancelled

  Shipping_Days
0             1
1             2
2             3
3             4
4             5

df.tail(5)

    Order_ID  Order_Date Customer_Name         Product        Category
\
46  ORD-1047  2025-01-27     Sana Khan  Cotton T-Shirt        Clothing

47  ORD-1048  2025-01-02   Rahul Verma       Rice 5 Kg       Groceries

48  ORD-1049  2025-01-05   Aarav Mehta  Ballpoint Pens      Stationery

49  ORD-1050  2025-01-08   Neha Sharma       Desk Lamp  Home & Kitchen

50  ORD-1036  2025-01-22   Rahul Verma   Running Shoes        Footwear


   Quantity Unit_Price Discount_Pct Payment_Method       City
Order_Status  \
46        1     569.05           12    Credit Card    Kolkata
Processing
```

```
47        5     418.95             0     Debit Card     Jaipur
Returned
48        2      99.00            25            COD  Hyderabad
Shipped
49        3    1079.10             5    Net Banking    Chennai
Cancelled
50        4    2199.00             8            UPI     Mumbai
Delivered

   Shipping_Days
46             5
47             6
48             7
49             1
50             1
```

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Number of rows: 51
Number of columns: 12
```

```
print(df.columns.tolist())
```

```
['Order_ID', 'Order_Date', 'Customer_Name', 'Product', 'Category',
'Quantity', 'Unit_Price', 'Discount_Pct', 'Payment_Method', 'City',
'Order_Status', 'Shipping_Days']
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 51 entries, 0 to 50
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order_ID        51 non-null     str
 1   Order_Date      50 non-null     str
 2   Customer_Name   51 non-null     str
 3   Product         51 non-null     str
 4   Category        50 non-null     str
 5   Quantity        49 non-null     str
 6   Unit_Price      50 non-null     str
 7   Discount_Pct    50 non-null     str
 8   Payment_Method  51 non-null     str
 9   City            50 non-null     str
 10  Order_Status    50 non-null     str
 11  Shipping_Days   51 non-null     str
dtypes: str(12)
memory usage: 9.1 KB
```

```
df.describe(include='all')
```

```
        Order_ID  Order_Date Customer_Name         Product
Category  \
count         51          50            51              51
50
unique        50          29            15              11
7
top     ORD-1036  2025-01-22   Neha Sharma  Wireless Mouse
Electronics
freq           2           3             5               6
16

       Quantity Unit_Price Discount_Pct Payment_Method       City  \
count        49         50           50             51         50
unique        8         36           11              6          9
top           2     799.00            5            UPI  Hyderabad
freq         14          3           10             11          7

       Order_Status Shipping_Days
count            50            51
unique            6             9
top       Delivered             1
freq             10             8
```

```
print("Unique Order IDs:", df['Order_ID'].nunique())
```

```
Unique Order IDs: 50
```

```
missing_values = df.isna().sum()

print("Missing values in each column:")
print(missing_values)

print("\nTotal missing values:", df.isna().sum().sum())
```

```
Missing values in each column:
Order_ID          0
Order_Date        1
Customer_Name     0
Product           0
Category          1
Quantity          2
Unit_Price        1
Discount_Pct      1
Payment_Method    0
City              1
Order_Status      1
Shipping_Days     0
dtype: int64

Total missing values: 8
```

```
print("Number of duplicate records:", df.duplicated().sum())

duplicate_rows = df[df.duplicated(keep=False)]

duplicate_rows
```

```
Number of duplicate records: 1
```

```
    Order_ID  Order_Date Customer_Name        Product  Category
Quantity  \
35  ORD-1036  2025-01-22   Rahul Verma  Running Shoes  Footwear
4
50  ORD-1036  2025-01-22   Rahul Verma  Running Shoes  Footwear
4

   Unit_Price Discount_Pct Payment_Method    City Order_Status
Shipping_Days
35    2199.00            8            UPI  Mumbai    Delivered
1
50    2199.00            8            UPI  Mumbai    Delivered
1
```

```
df_clean = df.drop_duplicates().copy()

print("Original dataset shape:", df.shape)
print("After removing duplicates:", df_clean.shape)

print("Remaining duplicates:", df_clean.duplicated().sum())
```

```
Original dataset shape: (51, 12)
After removing duplicates: (50, 12)
Remaining duplicates: 0
```

```
text_columns = [
    'Customer_Name',
    'Product',
    'Category',
    'Payment_Method',
    'City',
    'Order_Status'
]

for column in text_columns:
    df_clean[column] = df_clean[column].apply(
        lambda value: value.strip().title()
        if isinstance(value, str)
        else value
    )

df_clean.head()
```

```
   Order_ID  Order_Date Customer_Name            Product
Category  \
0  ORD-1001  2025-01-01   Neha Sharma     Wireless Mouse
Electronics
1  ORD-1002  2025-01-04   Neha Sharma  Bluetooth Speaker
Clothing
2  ORD-1003  2025-01-07    Ravi Kumar          Rice 5 Kg
Electronics
3  ORD-1004  2025-01-10    Isha Reddy     Ballpoint Pens
Stationery
4  ORD-1005  2025-01-13   Karan Patel          Desk Lamp  Home &
Kitchen

  Quantity Unit_Price Discount_Pct Payment_Method       City
Order_Status  \
0        1     799.00            0            Upi  Hyderabad
Delivered
1        2     569.05            5    Credit Card    Chennai
Processing
2        1     418.95           10     Debit Card  Bengaluru
Returned
3        3      99.00           15    Credit Card     Mumbai
Shipped
4        2    1079.10           20    Net Banking  Hyderabad
Cancelled

  Shipping_Days
0             1
1             2
2             3
3             4
4             5
```

```python
numeric_columns = [
    'Quantity',
    'Unit_Price',
    'Discount_Pct',
    'Shipping_Days'
]

for column in numeric_columns:
    df_clean[column] = pd.to_numeric(
        df_clean[column],
        errors='coerce'
    )

df_clean.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50 entries, 0 to 49
```

```
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   Order_ID        50 non-null     str    
 1   Order_Date      49 non-null     str    
 2   Customer_Name   50 non-null     str    
 3   Product         50 non-null     str    
 4   Category        49 non-null     str    
 5   Quantity        47 non-null     float64
 6   Unit_Price      47 non-null     float64
 7   Discount_Pct    48 non-null     float64
 8   Payment_Method  50 non-null     str    
 9   City            49 non-null     str    
 10  Order_Status    49 non-null     str    
 11  Shipping_Days   49 non-null     float64
dtypes: float64(4), str(8)
memory usage: 8.4 KB
```

```
print("Negative quantities:")
display(df_clean[df_clean['Quantity'] < 0])

print("Zero or negative prices:")
display(df_clean[df_clean['Unit_Price'] <= 0])

print("Invalid discount percentages:")
display(
    df_clean[
        (df_clean['Discount_Pct'] < 0) |
        (df_clean['Discount_Pct'] > 100)
    ]
)

print("Suspicious shipping days:")
display(df_clean[df_clean['Shipping_Days'] > 30])

print("Suspiciously high quantities:")
display(df_clean[df_clean['Quantity'] > 100])
```

```
Negative quantities:

    Order_ID  Order_Date Customer_Name         Product     Category  \
Quantity  \
18  ORD-1019  2025-01-27  Vikram Singh  Wireless Mouse  Electronics  
-3.0  

    Unit_Price  Discount_Pct Payment_Method       City Order_Status  \
18       799.0          25.0            Cod  Bengaluru      Shipped  

    Shipping_Days
18            5.0
```

```
Zero or negative prices:

    Order_ID  Order_Date Customer_Name         Product  Category
Quantity  \
19  ORD-1020  2025-01-02    Priya Nair  Cotton T-Shirt  Clothing
3.0

    Unit_Price  Discount_Pct Payment_Method    City Order_Status  \
19      -49.99           5.0    Net Banking  Mumbai    Cancelled

    Shipping_Days
19            6.0

Invalid discount percentages:

    Order_ID  Order_Date Customer_Name      Product     Category
Quantity  \
20  ORD-1021  2025-01-05     Arjun Das    Rice 5 Kg    Groceries
1.0
25  ORD-1026  2025-01-20   Neha Sharma  Usb-C Cable  Electronics
4.0

    Unit_Price  Discount_Pct Payment_Method     City Order_Status  \
20       399.0         150.0            Upi     Pune    Delivered
25       299.0         -10.0            Upi  Chennai    Delivered

    Shipping_Days
20            7.0
25            5.0

Suspicious shipping days:

    Order_ID  Order_Date Customer_Name     Product        Category
Quantity  \
24  ORD-1025  2025-01-17   Aarav Mehta  Coffee Mug  Home & Kitchen
2.0

    Unit_Price  Discount_Pct Payment_Method       City Order_Status  \
24       224.1          20.0    Net Banking  Hyderabad    Cancelled

    Shipping_Days
24          999.0

Suspiciously high quantities:

    Order_ID  Order_Date Customer_Name    Product        Category
Quantity  \
22  ORD-1023  2025-01-11     Sana Khan  Desk Lamp  Home & Kitchen
500.0

    Unit_Price  Discount_Pct Payment_Method     City Order_Status  \
```

```
22     1258.95          10.0     Debit Card  Kolkata     Returned

    Shipping_Days
22            2.0
```

```
print("Missing or invalid numeric values:")

for column in numeric_columns:
    print(f"{column}: {df_clean[column].isna().sum()}")
```

```
Missing or invalid numeric values:
Quantity: 3
Unit_Price: 3
Discount_Pct: 2
Shipping_Days: 1
```

```
print("Data cleaning summary")
print("---------------------")

print("Original records:", len(df))
print("Records after duplicate removal:", len(df_clean))
print("Duplicates removed:", len(df) - len(df_clean))

print("\nMissing values after conversion:")
print(df_clean.isna().sum())
```

```
Data cleaning summary
---------------------
Original records: 51
Records after duplicate removal: 50
Duplicates removed: 1

Missing values after conversion:
Order_ID          0
Order_Date        1
Customer_Name     0
Product           0
Category          1
Quantity          3
Unit_Price        3
Discount_Pct      2
Payment_Method    0
City              1
Order_Status      1
Shipping_Days     1
dtype: int64
```

```
df_clean['Order_Date'] = pd.to_datetime(
    df_clean['Order_Date'],
    errors='coerce'
)
```

```
print("Date data type:", df_clean['Order_Date'].dtype)

print("Invalid or missing dates:",
      df_clean['Order_Date'].isna().sum())
```

```
Date data type: datetime64[us]
Invalid or missing dates: 2
```

```
df_clean['Year'] = df_clean['Order_Date'].dt.year

df_clean['Month_Number'] = df_clean['Order_Date'].dt.month

df_clean['Day'] = df_clean['Order_Date'].dt.day

df_clean['Month'] = df_clean['Order_Date'].dt.strftime('%Y-%m')

df_clean[['Order_Date', 'Year', 'Month_Number', 'Day',
'Month']].head(10)
```

	Order_Date	Year	Month_Number	Day	Month
0	2025-01-01	2025.0	1.0	1.0	2025-01
1	2025-01-04	2025.0	1.0	4.0	2025-01
2	2025-01-07	2025.0	1.0	7.0	2025-01
3	2025-01-10	2025.0	1.0	10.0	2025-01
4	2025-01-13	2025.0	1.0	13.0	2025-01
5	2025-01-16	2025.0	1.0	16.0	2025-01
6	2025-01-19	2025.0	1.0	19.0	2025-01
7	2025-01-22	2025.0	1.0	22.0	2025-01
8	2025-01-25	2025.0	1.0	25.0	2025-01
9	NaT	NaN	NaN	NaN	NaN

```
df_clean = df_clean.sort_values(
    by='Order_Date',
    ascending=True
)

df_clean.head(10)
```

	Order_ID	Order_Date	Customer_Name	Product	Category
0	ORD-1001	2025-01-01	Neha Sharma	Wireless Mouse	Electronics
28	ORD-1029	2025-01-01	Ravi Kumar	Cotton T-Shirt	Clothing
47	ORD-1048	2025-01-02	Rahul Verma	Rice 5 Kg	Groceries
19	ORD-1020	2025-01-02	Priya Nair	Cotton T-Shirt	Clothing
38	ORD-1039	2025-01-03	Ravi Kumar	Rice 5 Kg	Groceries

```
1   ORD-1002 2025-01-04   Neha Sharma  Bluetooth Speaker     Clothing

29  ORD-1030 2025-01-04    Ananya Rao          Rice 5 Kg    Groceries

48  ORD-1049 2025-01-05   Aarav Mehta     Ballpoint Pens   Stationery

20  ORD-1021 2025-01-05     Arjun Das          Rice 5 Kg    Groceries

39  ORD-1040 2025-01-06    Isha Reddy     Ballpoint Pens   Stationery


    Quantity  Unit_Price  Discount_Pct Payment_Method       City
Order_Status  \
0        1.0      799.00           0.0            Upi  Hyderabad
Delivered
28       2.0      599.00          25.0            Cod       Pune
Shipped
47       5.0      418.95           0.0     Debit Card     Jaipur
Returned
19       3.0      -49.99           5.0    Net Banking     Mumbai
Cancelled
38       2.0      399.00          25.0            Cod    Kolkata
Shipped
1        2.0      569.05           5.0    Credit Card    Chennai
Processing
29       3.0      359.10           5.0    Net Banking      Delhi
Cancelled
48       2.0       99.00          25.0            Cod  Hyderabad
Shipped
20       1.0      399.00         150.0            Upi       Pune
Delivered
39       3.0       89.10           5.0    Net Banking     Jaipur
Cancelled

    Shipping_Days    Year  Month_Number  Day    Month
0             1.0  2025.0           1.0  1.0  2025-01
28            1.0  2025.0           1.0  1.0  2025-01
47            6.0  2025.0           1.0  2.0  2025-01
19            6.0  2025.0           1.0  2.0  2025-01
38            4.0  2025.0           1.0  3.0  2025-01
1             2.0  2025.0           1.0  4.0  2025-01
29            2.0  2025.0           1.0  4.0  2025-01
48            7.0  2025.0           1.0  5.0  2025-01
20            7.0  2025.0           1.0  5.0  2025-01
39            5.0  2025.0           1.0  6.0  2025-01
```

```python
df_clean['Revenue'] = (
    df_clean['Quantity'] *
    df_clean['Unit_Price'] *
    (1 - df_clean['Discount_Pct'] / 100)
```

```
)

# Prevent revenue calculations for invalid input values
invalid_revenue = (
    df_clean['Quantity'].isna() |
    (df_clean['Quantity'] <= 0) |
    df_clean['Unit_Price'].isna() |
    (df_clean['Unit_Price'] <= 0) |
    df_clean['Discount_Pct'].isna() |
    (df_clean['Discount_Pct'] < 0) |
    (df_clean['Discount_Pct'] > 100)
)

df_clean.loc[invalid_revenue, 'Revenue'] = np.nan

df_clean[['Order_ID', 'Quantity', 'Unit_Price',
          'Discount_Pct', 'Revenue']].head(10)
```

	Order_ID	Quantity	Unit_Price	Discount_Pct	Revenue
0	ORD-1001	1.0	799.00	0.0	799.000
28	ORD-1029	2.0	599.00	25.0	898.500
47	ORD-1048	5.0	418.95	0.0	2094.750
19	ORD-1020	3.0	-49.99	5.0	NaN
38	ORD-1039	2.0	399.00	25.0	598.500
1	ORD-1002	2.0	569.05	5.0	1081.195
29	ORD-1030	3.0	359.10	5.0	1023.435
48	ORD-1049	2.0	99.00	25.0	148.500
20	ORD-1021	1.0	399.00	150.0	NaN
39	ORD-1040	3.0	89.10	5.0	253.935

```
df_clean[
    ['Order_ID', 'Quantity', 'Unit_Price',
     'Discount_Pct', 'Revenue']
].dropna().head(10)
```

	Order_ID	Quantity	Unit_Price	Discount_Pct	Revenue
0	ORD-1001	1.0	799.00	0.0	799.000
28	ORD-1029	2.0	599.00	25.0	898.500
47	ORD-1048	5.0	418.95	0.0	2094.750
38	ORD-1039	2.0	399.00	25.0	598.500
1	ORD-1002	2.0	569.05	5.0	1081.195
29	ORD-1030	3.0	359.10	5.0	1023.435
48	ORD-1049	2.0	99.00	25.0	148.500
39	ORD-1040	3.0	89.10	5.0	253.935
11	ORD-1012	2.0	379.05	5.0	720.195
2	ORD-1003	1.0	418.95	10.0	377.055

```
print("Revenue data type:", df_clean['Revenue'].dtype)

print("Missing Revenue values:",
      df_clean['Revenue'].isna().sum())
```

```
print("Total Revenue:",
      df_clean['Revenue'].sum())
```

```
Revenue data type: float64
Missing Revenue values: 12
Total Revenue: 647248.9475
```

```
sorted_df = df_clean.sort_values(
    by='Revenue',
    ascending=False
)

sorted_df.head(5)
```

```
    Order_ID Order_Date Customer_Name            Product
Category  \
22  ORD-1023 2025-01-11     Sana Khan          Desk Lamp  Home &
Kitchen
23  ORD-1024 2025-01-14   Rahul Verma  Bluetooth Speaker
Electronics
35  ORD-1036 2025-01-22   Rahul Verma      Running Shoes
Footwear
5   ORD-1006 2025-01-16    Ananya Rao  Bluetooth Speaker
Electronics
44  ORD-1045 2025-01-21     Arjun Das      Running Shoes
Footwear

    Quantity  Unit_Price  Discount_Pct Payment_Method     City
Order_Status  \
22     500.0     1258.95          10.0     Debit Card  Kolkata
Returned
23       3.0     9999.99          15.0            Cod   Jaipur
Shipped
35       4.0     2199.00           8.0            Upi   Mumbai
Delivered
5        4.0     1499.00           8.0            Upi    Delhi
Shipped
44       2.0     1979.10          20.0    Net Banking     Pune
Cancelled

    Shipping_Days    Year  Month_Number   Day    Month      Revenue
22            2.0  2025.0           1.0  11.0  2025-01  566527.5000
23            3.0  2025.0           1.0  14.0  2025-01   25499.9745
35            1.0  2025.0           1.0  22.0  2025-01    8092.3200
5             6.0  2025.0           1.0  16.0  2025-01    5516.3200
44            3.0  2025.0           1.0  21.0  2025-01    3166.5600
```

```
selected_columns = df_clean[
    ['Order_ID', 'Product', 'Quantity',
     'Unit_Price', 'Discount_Pct', 'Revenue']
```

```
]

selected_columns.head(10)
```

```
    Order_ID            Product  Quantity  Unit_Price  Discount_Pct
Revenue
0   ORD-1001     Wireless Mouse       1.0      799.00           0.0
799.000
28  ORD-1029     Cotton T-Shirt       2.0      599.00          25.0
898.500
47  ORD-1048          Rice 5 Kg       5.0      418.95           0.0
2094.750
19  ORD-1020     Cotton T-Shirt       3.0      -49.99           5.0
NaN
38  ORD-1039          Rice 5 Kg       2.0      399.00          25.0
598.500
1   ORD-1002  Bluetooth Speaker       2.0      569.05           5.0
1081.195
29  ORD-1030          Rice 5 Kg       3.0      359.10           5.0
1023.435
48  ORD-1049     Ballpoint Pens       2.0       99.00          25.0
148.500
20  ORD-1021          Rice 5 Kg       1.0      399.00         150.0
NaN
39  ORD-1040     Ballpoint Pens       3.0       89.10           5.0
253.935
```

```
high_revenue_orders = df_clean[
    df_clean['Revenue'] > 500
]

high_revenue_orders.head(10)
```

```
    Order_ID Order_Date Customer_Name            Product
Category  \
0   ORD-1001 2025-01-01   Neha Sharma     Wireless Mouse
Electronics
28  ORD-1029 2025-01-01    Ravi Kumar     Cotton T-Shirt
Clothing
47  ORD-1048 2025-01-02   Rahul Verma          Rice 5 Kg
Groceries
38  ORD-1039 2025-01-03    Ravi Kumar          Rice 5 Kg
Groceries
1   ORD-1002 2025-01-04   Neha Sharma  Bluetooth Speaker
Clothing
29  ORD-1030 2025-01-04    Ananya Rao          Rice 5 Kg
Groceries
11  ORD-1012 2025-01-06                        Rice 5 Kg
Groceries
49  ORD-1050 2025-01-08   Neha Sharma          Desk Lamp  Home &
```

```
Kitchen
40  ORD-1041 2025-01-09   Karan Patel          Desk Lamp  Home &
Kitchen
31  ORD-1032 2025-01-10    Priya Nair          Desk Lamp  Home &
Kitchen

    Quantity  Unit_Price  Discount_Pct Payment_Method       City
Order_Status  \
0        1.0      799.00           0.0            Upi  Hyderabad
Delivered
28       2.0      599.00          25.0            Cod       Pune
Shipped
47       5.0      418.95           0.0     Debit Card     Jaipur
Returned
38       2.0      399.00          25.0            Cod    Kolkata
Shipped
1        2.0      569.05           5.0    Credit Card    Chennai
Processing
29       3.0      359.10           5.0    Net Banking      Delhi
Cancelled
11       2.0      379.05           5.0    Credit Card     Mumbai
Processing
49       3.0     1079.10           5.0    Net Banking    Chennai
Cancelled
40       1.0     1199.00           0.0            Upi  Hyderabad
Delivered
31       2.0     1139.05           5.0    Credit Card     Jaipur
Processing

    Shipping_Days    Year  Month_Number   Day    Month   Revenue
0             1.0  2025.0           1.0   1.0  2025-01   799.000
28            1.0  2025.0           1.0   1.0  2025-01   898.500
47            6.0  2025.0           1.0   2.0  2025-01  2094.750
38            4.0  2025.0           1.0   3.0  2025-01   598.500
1             2.0  2025.0           1.0   4.0  2025-01  1081.195
29            2.0  2025.0           1.0   4.0  2025-01  1023.435
11            5.0  2025.0           1.0   6.0  2025-01   720.195
49            1.0  2025.0           1.0   8.0  2025-01  3075.435
40            6.0  2025.0           1.0   9.0  2025-01  1199.000
31            4.0  2025.0           1.0  10.0  2025-01  2164.195
```

```
high_discount_orders = df_clean[
    df_clean['Discount_Pct'] > 10
]

high_discount_orders[
    ['Order_ID', 'Product', 'Discount_Pct']
]
```

	Order_ID	Product	Discount_Pct
28	ORD-1029	Cotton T-Shirt	25.0
38	ORD-1039	Rice 5 Kg	25.0
48	ORD-1049	Ballpoint Pens	25.0
20	ORD-1021	Rice 5 Kg	150.0
3	ORD-1004	Ballpoint Pens	15.0
13	ORD-1014	Desk Lamp	15.0
4	ORD-1005	Desk Lamp	20.0
23	ORD-1024	Bluetooth Speaker	15.0
14	ORD-1015	Bluetooth Speaker	20.0
33	ORD-1034	Coffee Mug	15.0
24	ORD-1025	Coffee Mug	20.0
43	ORD-1044	Usb-C Cable	15.0
34	ORD-1035	Usb-C Cable	20.0
6	ORD-1007	Coffee Mug	12.0
16	ORD-1017	Usb-C Cable	12.0
44	ORD-1045	Running Shoes	20.0
26	ORD-1027	Running Shoes	12.0
36	ORD-1037	Wireless Mouse	12.0
46	ORD-1047	Cotton T-Shirt	12.0
18	ORD-1019	Wireless Mouse	25.0

```
high_quantity_orders = df_clean[
    df_clean['Quantity'] > 2
]

high_quantity_orders[
    ['Order_ID', 'Product', 'Quantity']
]
```

	Order_ID	Product	Quantity
47	ORD-1048	Rice 5 Kg	5.0
19	ORD-1020	Cotton T-Shirt	3.0
29	ORD-1030	Rice 5 Kg	3.0
39	ORD-1040	Ballpoint Pens	3.0
49	ORD-1050	Desk Lamp	3.0
3	ORD-1004	Ballpoint Pens	3.0
22	ORD-1023	Desk Lamp	500.0
13	ORD-1014	Desk Lamp	3.0
23	ORD-1024	Bluetooth Speaker	3.0
33	ORD-1034	Coffee Mug	3.0
5	ORD-1006	Bluetooth Speaker	4.0
43	ORD-1044	Usb-C Cable	3.0
25	ORD-1026	Usb-C Cable	4.0
35	ORD-1036	Running Shoes	4.0
7	ORD-1008	Usb-C Cable	5.0
45	ORD-1046	Wireless Mouse	4.0
17	ORD-1018	Running Shoes	5.0
37	ORD-1038	Cotton T-Shirt	5.0
9	ORD-1010	Wireless Mouse	3.0

```
filtered_orders = df_clean[
    (df_clean['Quantity'] > 2) &
    (df_clean['Revenue'] > 500)
]

filtered_orders[
    ['Order_ID', 'Product', 'Quantity', 'Revenue']
]
```

	Order_ID	Product	Quantity	Revenue
47	ORD-1048	Rice 5 Kg	5.0	2094.7500
29	ORD-1030	Rice 5 Kg	3.0	1023.4350
49	ORD-1050	Desk Lamp	3.0	3075.4350
22	ORD-1023	Desk Lamp	500.0	566527.5000
13	ORD-1014	Desk Lamp	3.0	3057.4500
23	ORD-1024	Bluetooth Speaker	3.0	25499.9745
33	ORD-1034	Coffee Mug	3.0	634.9500
5	ORD-1006	Bluetooth Speaker	4.0	5516.3200
43	ORD-1044	Usb-C Cable	3.0	762.4500
35	ORD-1036	Running Shoes	4.0	8092.3200
45	ORD-1046	Wireless Mouse	4.0	2940.3200
37	ORD-1038	Cotton T-Shirt	5.0	3144.7500
9	ORD-1010	Wireless Mouse	3.0	2049.4350

```
filtered_orders_or = df_clean[
    (df_clean['City'] == 'Hyderabad') |
    (df_clean['City'] == 'Chennai')
]

filtered_orders_or[
    ['Order_ID', 'Product', 'City', 'Revenue']
]
```

	Order_ID	Product	City	Revenue
0	ORD-1001	Wireless Mouse	Hyderabad	799.000
1	ORD-1002	Bluetooth Speaker	Chennai	1081.195
48	ORD-1049	Ballpoint Pens	Hyderabad	148.500
49	ORD-1050	Desk Lamp	Chennai	3075.435
40	ORD-1041	Desk Lamp	Hyderabad	1199.000
41	ORD-1042	Bluetooth Speaker	Chennai	2705.695
4	ORD-1005	Desk Lamp	Hyderabad	1726.560
32	ORD-1033	Bluetooth Speaker	Hyderabad	1416.555
33	ORD-1034	Coffee Mug	Chennai	634.950
24	ORD-1025	Coffee Mug	Hyderabad	358.560
25	ORD-1026	Usb-C Cable	Chennai	NaN
16	ORD-1017	Usb-C Cable	Hyderabad	NaN
17	ORD-1018	Running Shoes	Chennai	NaN
8	ORD-1009	Running Shoes	Hyderabad	NaN
9	ORD-1010	Wireless Mouse	Chennai	2049.435

```
df_clean.loc[
    df_clean['Revenue'] > 500,
    ['Order_ID', 'Product', 'Revenue']
].head(10)
```

```
    Order_ID            Product   Revenue
0   ORD-1001     Wireless Mouse   799.000
28  ORD-1029     Cotton T-Shirt   898.500
47  ORD-1048          Rice 5 Kg  2094.750
38  ORD-1039          Rice 5 Kg   598.500
1   ORD-1002  Bluetooth Speaker  1081.195
29  ORD-1030          Rice 5 Kg  1023.435
11  ORD-1012          Rice 5 Kg   720.195
49  ORD-1050          Desk Lamp  3075.435
40  ORD-1041          Desk Lamp  1199.000
31  ORD-1032          Desk Lamp  2164.195
```

```
# First 5 rows and first 4 columns

df_clean.iloc[0:5, 0:4]
```

```
    Order_ID Order_Date Customer_Name         Product
0   ORD-1001 2025-01-01   Neha Sharma  Wireless Mouse
28  ORD-1029 2025-01-01    Ravi Kumar  Cotton T-Shirt
47  ORD-1048 2025-01-02   Rahul Verma       Rice 5 Kg
19  ORD-1020 2025-01-02    Priya Nair  Cotton T-Shirt
38  ORD-1039 2025-01-03    Ravi Kumar       Rice 5 Kg
```

```
df_clean = df_clean.reset_index(drop=True)

df_clean.head()
```

```
   Order_ID Order_Date Customer_Name         Product     Category
Quantity  \
0  ORD-1001 2025-01-01   Neha Sharma  Wireless Mouse  Electronics
1.0
1  ORD-1029 2025-01-01    Ravi Kumar  Cotton T-Shirt     Clothing
2.0
2  ORD-1048 2025-01-02   Rahul Verma       Rice 5 Kg    Groceries
5.0
3  ORD-1020 2025-01-02    Priya Nair  Cotton T-Shirt     Clothing
3.0
4  ORD-1039 2025-01-03    Ravi Kumar       Rice 5 Kg    Groceries
2.0

   Unit_Price  Discount_Pct Payment_Method       City Order_Status  \
0      799.00           0.0            Upi  Hyderabad    Delivered
1      599.00          25.0            Cod       Pune      Shipped
2      418.95           0.0     Debit Card     Jaipur     Returned
3      -49.99           5.0    Net Banking     Mumbai    Cancelled
4      399.00          25.0            Cod    Kolkata      Shipped
```

```
   Shipping_Days    Year  Month_Number  Day    Month  Revenue
0            1.0  2025.0           1.0  1.0  2025-01   799.00
1            1.0  2025.0           1.0  1.0  2025-01   898.50
2            6.0  2025.0           1.0  2.0  2025-01  2094.75
3            6.0  2025.0           1.0  2.0  2025-01      NaN
4            4.0  2025.0           1.0  3.0  2025-01   598.50
```

```
region_summary = df_clean.groupby('City').agg(
    TotalRevenue=('Revenue', 'sum'),
    Orders=('Order_ID', 'nunique'),
    AverageRevenuePerOrder=('Revenue', 'mean')
)

region_summary = region_summary.sort_values(
    by='TotalRevenue',
    ascending=False
)

region_summary
```

```
           TotalRevenue  Orders  AverageRevenuePerOrder
City
Kolkata     569884.3240       6           113976.864800
Jaipur       30012.8545       6             7503.213625
Delhi        12803.5200       5             2560.704000
Mumbai        9827.4150       6             2456.853750
Chennai       9546.7100       7             1909.342000
Hyderabad     5648.1750       8              941.362500
Pune          4826.5790       5             1206.644750
Bengaluru     1641.9200       6              410.480000
```

```
highest_revenue_city = region_summary['TotalRevenue'].idxmax()

print("Highest revenue city:", highest_revenue_city)

print(
    "Total revenue:",
    region_summary.loc[highest_revenue_city, 'TotalRevenue']
)
```

```
Highest revenue city: Kolkata
Total revenue: 569884.324
```

```
product_summary = df_clean.groupby('Product').agg(
    TotalRevenue=('Revenue', 'sum'),
    UnitsSold=('Quantity', 'sum'),
    Orders=('Order_ID', 'nunique')
)

product_summary = product_summary.sort_values(
```

```
    by='TotalRevenue',
    ascending=False
)

product_summary
```

Product	TotalRevenue	UnitsSold	Orders
Desk Lamp	577750.1400	511.0	6
Bluetooth Speaker	38378.2995	14.0	6
Running Shoes	11258.8800	14.0	5
Wireless Mouse	6456.7190	6.0	6
Cotton T-Shirt	5143.0140	12.0	5
Rice 5 Kg	4813.9350	14.0	6
Coffee Mug	1228.8150	6.0	5
Usb-C Cable	1193.0100	15.0	5
Ballpoint Pens	1026.1350	12.0	6

```
highest_revenue_product = product_summary['TotalRevenue'].idxmax()

print("Highest revenue product:", highest_revenue_product)

print(
    "Total revenue:",
    product_summary.loc[highest_revenue_product, 'TotalRevenue']
)
```

```
Highest revenue product: Desk Lamp
Total revenue: 577750.14
```

```
monthly_summary = df_clean.groupby('Month').agg(
    TotalRevenue=('Revenue', 'sum'),
    Orders=('Order_ID', 'nunique')
)

monthly_summary = monthly_summary.sort_index()

monthly_summary
```

Month	TotalRevenue	Orders
2025-01	644600.5125	48

```
highest_revenue_month = monthly_summary['TotalRevenue'].idxmax()

print("Highest revenue month:", highest_revenue_month)

print(
    "Total revenue:",
    monthly_summary.loc[highest_revenue_month, 'TotalRevenue']
)
```

```
Highest revenue month: 2025-01
Total revenue: 644600.5125000001
```

```
highest_revenue_orders = (
    df_clean
    .dropna(subset=['Revenue'])
    .sort_values(by='Revenue', ascending=False)
    [[
        'Order_ID',
        'Order_Date',
        'City',
        'Product',
        'Quantity',
        'Unit_Price',
        'Discount_Pct',
        'Revenue'
    ]]
    .head(5)
    .reset_index(drop=True)
)

highest_revenue_orders
```

```
   Order_ID Order_Date     City            Product  Quantity
Unit_Price  \
0  ORD-1023 2025-01-11  Kolkata          Desk Lamp     500.0
1258.95
1  ORD-1024 2025-01-14   Jaipur  Bluetooth Speaker       3.0
9999.99
2  ORD-1036 2025-01-22   Mumbai      Running Shoes       4.0
2199.00
3  ORD-1006 2025-01-16    Delhi  Bluetooth Speaker       4.0
1499.00
4  ORD-1045 2025-01-21     Pune      Running Shoes       2.0
1979.10

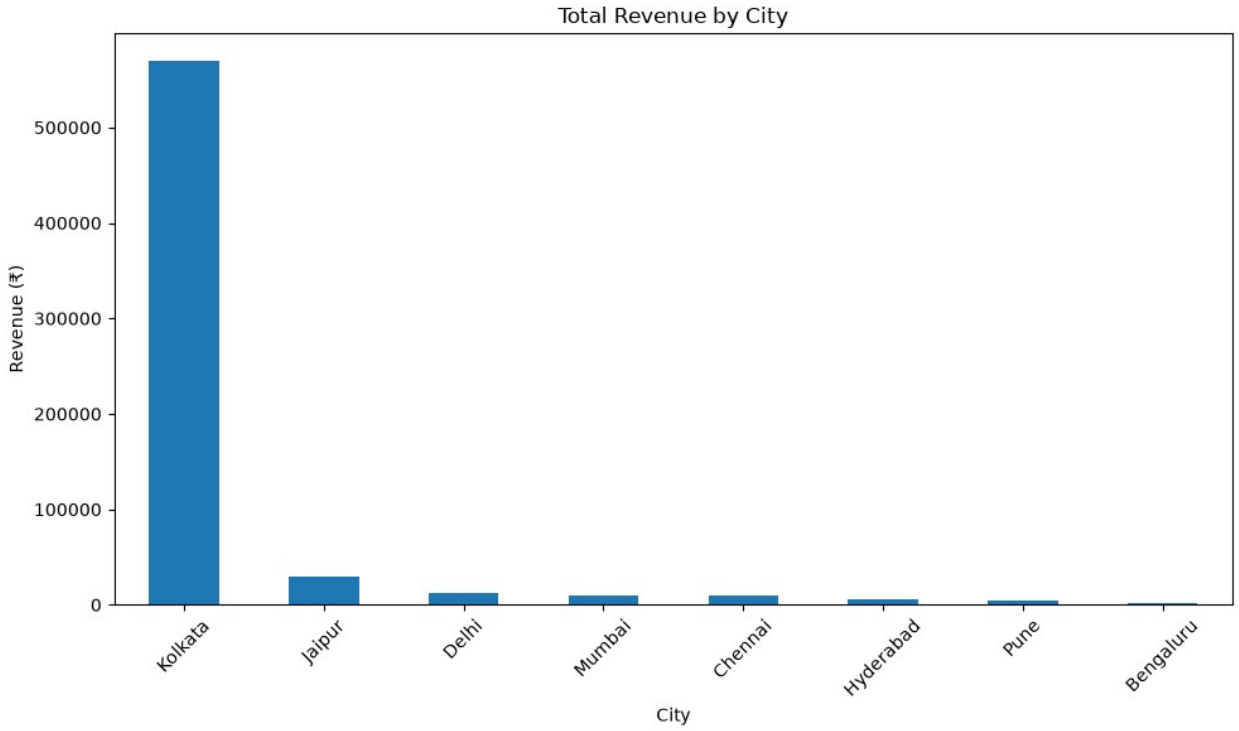
   Discount_Pct      Revenue
0          10.0  566527.5000
1          15.0   25499.9745
2           8.0    8092.3200
3           8.0    5516.3200
4          20.0    3166.5600
```

```
plt.figure(figsize=(10, 6))

region_summary['TotalRevenue'].plot(
    kind='bar',
    title='Total Revenue by City'
)
```

```
plt.xlabel('City')
plt.ylabel('Revenue (₹)')
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(10, 6))

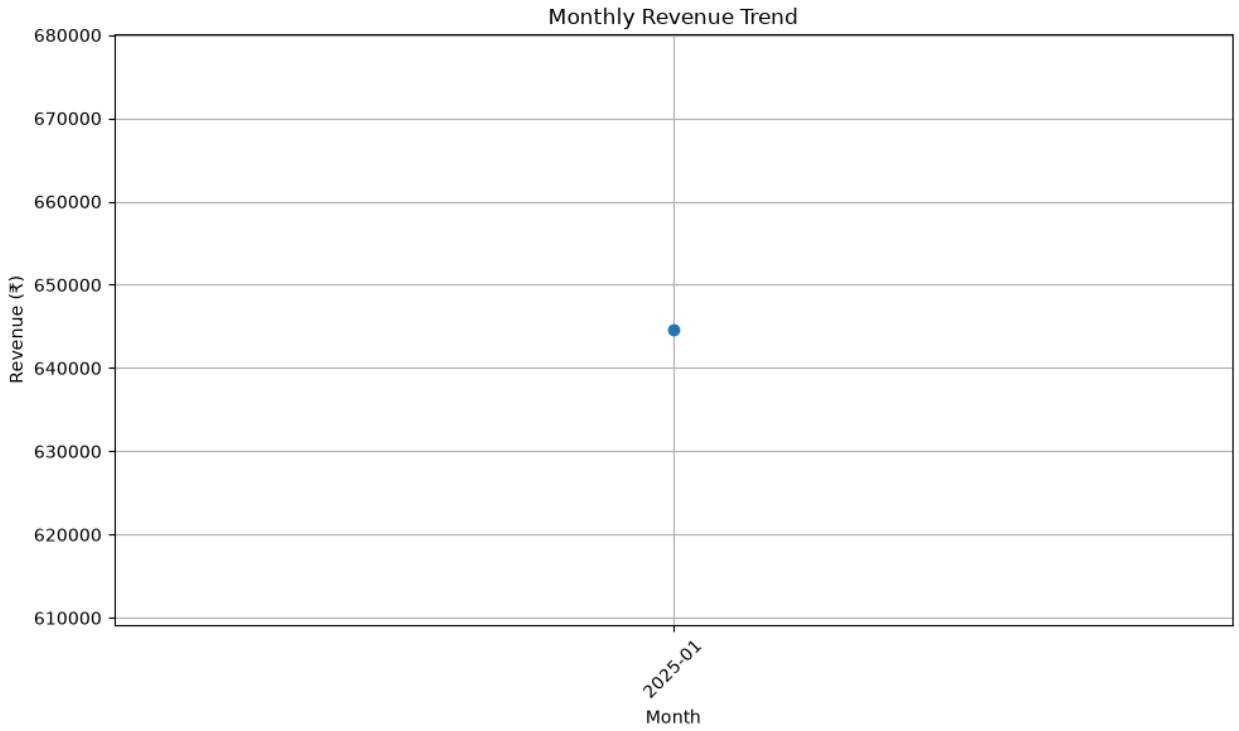
plt.plot(
    monthly_summary.index,
    monthly_summary['TotalRevenue'],
    marker='o',
    linestyle='-'
)

plt.title('Monthly Revenue Trend')
plt.xlabel('Month')
plt.ylabel('Revenue (₹)')

plt.xticks(rotation=45)

plt.grid(True)

plt.tight_layout()
plt.show()
```



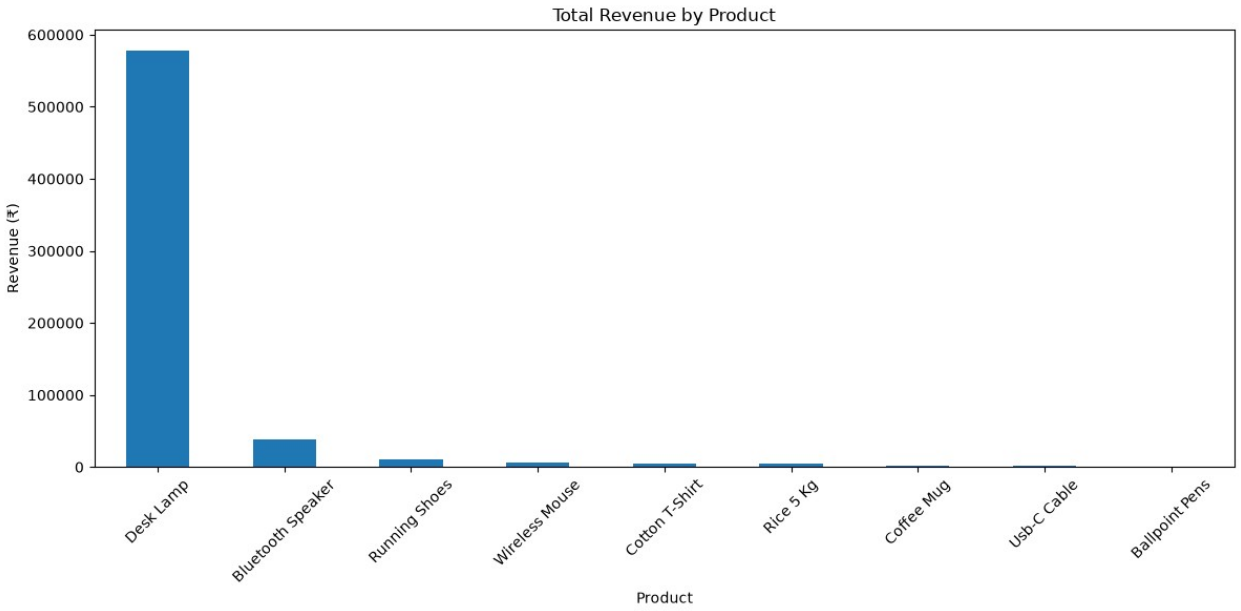

```
plt.figure(figsize=(12, 6))

product_summary['TotalRevenue'].plot(
    kind='bar',
    title='Total Revenue by Product'
)

plt.xlabel('Product')
plt.ylabel('Revenue (₹)')

plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```




```
print("Orders by City:")
display(df_clean['City'].value_counts())

print("Orders by Product:")
display(df_clean['Product'].value_counts())

print("Orders by Payment Method:")
display(df_clean['Payment_Method'].value_counts())
```

```
Orders by City:

City
Hyderabad    8
Chennai      7
Jaipur       6
Mumbai       6
Kolkata      6
Bengaluru    6
Pune         5
Delhi        5
Name: count, dtype: int64

Orders by Product:

Product
Wireless Mouse       6
Rice 5 Kg            6
Bluetooth Speaker    6
Ballpoint Pens       6
Desk Lamp            6
Cotton T-Shirt       5
Coffee Mug           5
```

```
Usb-C Cable          5
Running Shoes        5
Name: count, dtype: int64

Orders by Payment Method:

Payment_Method
Credit Card    11
Upi            10
Debit Card     10
Net Banking    10
Cod             9
Name: count, dtype: int64
```

```
print("Unique Cities:")
print(df_clean['City'].unique())

print("\nUnique Products:")
print(df_clean['Product'].unique())

print("\nUnique Categories:")
print(df_clean['Category'].unique())
```

```
Unique Cities:
<ArrowStringArray>
['Hyderabad',      'Pune',    'Jaipur',    'Mumbai',   'Kolkata',
'Chennai',
     'Delhi', 'Bengaluru',         nan]
Length: 9, dtype: str

Unique Products:
<ArrowStringArray>
[   'Wireless Mouse',    'Cotton T-Shirt',         'Rice 5 Kg',
 'Bluetooth Speaker',    'Ballpoint Pens',         'Desk Lamp',
        'Coffee Mug',       'Usb-C Cable',     'Running Shoes']
Length: 9, dtype: str

Unique Categories:
<ArrowStringArray>
[   'Electronics',       'Clothing',      'Groceries',
'Stationery',
 'Home & Kitchen',              nan,       'Footwear']
Length: 7, dtype: str
```

```
product_statistics = df_clean.groupby('Product').agg({
    'Revenue': ['sum', 'mean', 'min', 'max'],
    'Quantity': ['sum', 'mean'],
    'Unit_Price': ['min', 'max']
})

product_statistics
```

```
                          Revenue
Quantity  \
                              sum          mean       min           max
sum
Product

Ballpoint Pens         1026.1350    171.02250    93.555     253.9350
12.0
Bluetooth Speaker     38378.2995   6396.38325  1081.195   25499.9745
14.0
Coffee Mug             1228.8150    409.60500   235.305     634.9500
6.0
Cotton T-Shirt         5143.0140   1285.75350   500.764    3144.7500
12.0
Desk Lamp            577750.1400  96291.69000  1199.000  566527.5000
511.0
Rice 5 Kg              4813.9350    962.78700   377.055    2094.7500
14.0
Running Shoes         11258.8800   5629.44000  3166.560    8092.3200
14.0
Usb-C Cable            1193.0100    596.50500   430.560     762.4500
15.0
Wireless Mouse         6456.7190   1614.17975   667.964    2940.3200
6.0
```

```
                            Unit_Price
                          mean        min      max
Product
Ballpoint Pens        2.000000      89.10   103.95
Bluetooth Speaker     2.333333     569.05  9999.99
Coffee Mug            2.000000     224.10   261.45
Cotton T-Shirt        2.400000     -49.99   628.95
Desk Lamp            85.166667    1079.10  1258.95
Rice 5 Kg             2.333333     359.10   418.95
Running Shoes         2.800000    1979.10  2308.95
Usb-C Cable           3.000000     269.10   299.00
Wireless Mouse        1.200000     719.10   838.95
```

```
city_product_summary = df_clean.groupby(
    ['City', 'Product']
).agg(
    TotalRevenue=('Revenue', 'sum'),
    UnitsSold=('Quantity', 'sum'),
    TotalOrders=('Order_ID', 'nunique')
)

city_product_summary
```

```
                              TotalRevenue  UnitsSold  TotalOrders
City      Product
```

```
Bengaluru Coffee Mug             235.3050        1.0             1
          Cotton T-Shirt         599.0000        1.0             1
          Rice 5 Kg              377.0550        1.0             1
          Running Shoes            0.0000        1.0             1
          Usb-C Cable            430.5600        2.0             1
          Wireless Mouse           0.0000       -3.0             1
Chennai   Bluetooth Speaker     3786.8900        4.0             2
          Coffee Mug             634.9500        3.0             1
          Desk Lamp             3075.4350        3.0             1
          Running Shoes            0.0000        5.0             1
          Usb-C Cable              0.0000        4.0             1
          Wireless Mouse        2049.4350        3.0             1
Delhi     Ballpoint Pens         178.6950        2.0             1
          Bluetooth Speaker     5516.3200        4.0             1
          Cotton T-Shirt        3144.7500        5.0             1
          Rice 5 Kg             1023.4350        3.0             1
          Wireless Mouse        2940.3200        4.0             1
Hyderabad Ballpoint Pens         148.5000        2.0             1
          Bluetooth Speaker     1416.5550        1.0             1
          Coffee Mug             358.5600        2.0             1
          Desk Lamp             2925.5600        3.0             2
          Running Shoes            0.0000        2.0             1
          Usb-C Cable              0.0000        1.0             1
          Wireless Mouse         799.0000        1.0             1
Jaipur    Ballpoint Pens         253.9350        3.0             1
          Bluetooth Speaker    25499.9745        3.0             1
          Coffee Mug               0.0000        0.0             1
          Desk Lamp             2164.1950        2.0             1
          Rice 5 Kg             2094.7500        5.0             1
          Usb-C Cable              0.0000        5.0             1
Kolkata   Ballpoint Pens           99.0000        1.0             1
          Bluetooth Speaker     2158.5600        2.0             1
          Coffee Mug               0.0000        0.0             1
          Cotton T-Shirt         500.7640        1.0             1
          Desk Lamp           566527.5000      500.0             1
          Rice 5 Kg              598.5000        2.0             1
Mumbai    Ballpoint Pens         252.4500        3.0             1
          Cotton T-Shirt           0.0000        3.0             1
          Rice 5 Kg              720.1950        2.0             1
          Running Shoes         8092.3200        4.0             1
          Usb-C Cable            762.4500        3.0             1
          Wireless Mouse           0.0000        0.0             1
Pune      Ballpoint Pens          93.5550        1.0             1
          Cotton T-Shirt         898.5000        2.0             1
          Rice 5 Kg                0.0000        1.0             1
          Running Shoes         3166.5600        2.0             1
          Wireless Mouse         667.9640        1.0             1

# Convert grouped index into a normal column
```

```
city_product_reset = city_product_summary.reset_index()

display(city_product_reset.head())

# Sort monthly summary by its index

monthly_sorted = monthly_summary.sort_index()

display(monthly_sorted)
```

```
        City         Product  TotalRevenue  UnitsSold  TotalOrders
0  Bengaluru      Coffee Mug       235.305        1.0            1
1  Bengaluru  Cotton T-Shirt       599.000        1.0            1
2  Bengaluru       Rice 5 Kg       377.055        1.0            1
3  Bengaluru   Running Shoes         0.000        1.0            1
4  Bengaluru     Usb-C Cable       430.560        2.0            1
```

```
         TotalRevenue  Orders
Month
2025-01   644600.5125      48
```

```
renamed_df = df_clean.rename(columns={
    'Order_ID': 'OrderID',
    'Order_Date': 'Date',
    'Unit_Price': 'UnitPrice',
    'Discount_Pct': 'DiscountPercentage'
})

renamed_df.head()
```

```
    OrderID       Date Customer_Name         Product     Category
Quantity  \
0  ORD-1001 2025-01-01   Neha Sharma  Wireless Mouse  Electronics
1.0
1  ORD-1029 2025-01-01    Ravi Kumar  Cotton T-Shirt     Clothing
2.0
2  ORD-1048 2025-01-02   Rahul Verma       Rice 5 Kg    Groceries
5.0
3  ORD-1020 2025-01-02    Priya Nair  Cotton T-Shirt     Clothing
3.0
4  ORD-1039 2025-01-03    Ravi Kumar       Rice 5 Kg    Groceries
2.0

   UnitPrice  DiscountPercentage Payment_Method       City
Order_Status  \
0     799.00                 0.0            Upi  Hyderabad
Delivered
1     599.00                25.0            Cod       Pune
Shipped
2     418.95                 0.0     Debit Card     Jaipur
Returned
```

```
3     -49.99                  5.0    Net Banking     Mumbai
Cancelled
4     399.00                 25.0            Cod    Kolkata
Shipped

   Shipping_Days    Year  Month_Number  Day    Month  Revenue
0            1.0  2025.0           1.0  1.0  2025-01   799.00
1            1.0  2025.0           1.0  1.0  2025-01   898.50
2            6.0  2025.0           1.0  2.0  2025-01  2094.75
3            6.0  2025.0           1.0  2.0  2025-01      NaN
4            4.0  2025.0           1.0  3.0  2025-01   598.50
```

```
reduced_df = df_clean.copy()

reduced_df = reduced_df.drop(
    columns=['Customer_Name', 'Payment_Method']
)

reduced_df.head()
```

```
   Order_ID Order_Date         Product     Category  Quantity
Unit_Price  \
0  ORD-1001 2025-01-01  Wireless Mouse  Electronics       1.0
799.00
1  ORD-1029 2025-01-01  Cotton T-Shirt     Clothing       2.0
599.00
2  ORD-1048 2025-01-02       Rice 5 Kg    Groceries       5.0
418.95
3  ORD-1020 2025-01-02  Cotton T-Shirt     Clothing       3.0      -
49.99
4  ORD-1039 2025-01-03       Rice 5 Kg    Groceries       2.0
399.00

   Discount_Pct       City Order_Status  Shipping_Days    Year
Month_Number  \
0           0.0  Hyderabad    Delivered            1.0  2025.0
1.0
1          25.0       Pune      Shipped            1.0  2025.0
1.0
2           0.0     Jaipur     Returned            6.0  2025.0
1.0
3           5.0     Mumbai    Cancelled            6.0  2025.0
1.0
4          25.0    Kolkata      Shipped            4.0  2025.0
1.0

   Day    Month  Revenue
0  1.0  2025-01   799.00
1  1.0  2025-01   898.50
2  2.0  2025-01  2094.75
```

```
3  2.0  2025-01      NaN
4  3.0  2025-01   598.50

numeric_df = df_clean[
    ['Quantity', 'Unit_Price', 'Discount_Pct',
     'Shipping_Days', 'Revenue']
]

correlation_matrix = numeric_df.corr()

correlation_matrix
```

```
                Quantity  Unit_Price  Discount_Pct  Shipping_Days
Revenue
Quantity        1.000000    0.031350     -0.025538      -0.024750
0.999051
Unit_Price      0.031350    1.000000     -0.008065      -0.076183
0.074660
Discount_Pct   -0.025538   -0.008065      1.000000       0.051811 -
0.005571
Shipping_Days -0.024750   -0.076183      0.051811       1.000000 -
0.033087
Revenue         0.999051    0.074660     -0.005571      -0.033087
1.000000
```

```
plt.figure(figsize=(8, 6))

plt.imshow(correlation_matrix, cmap='coolwarm')

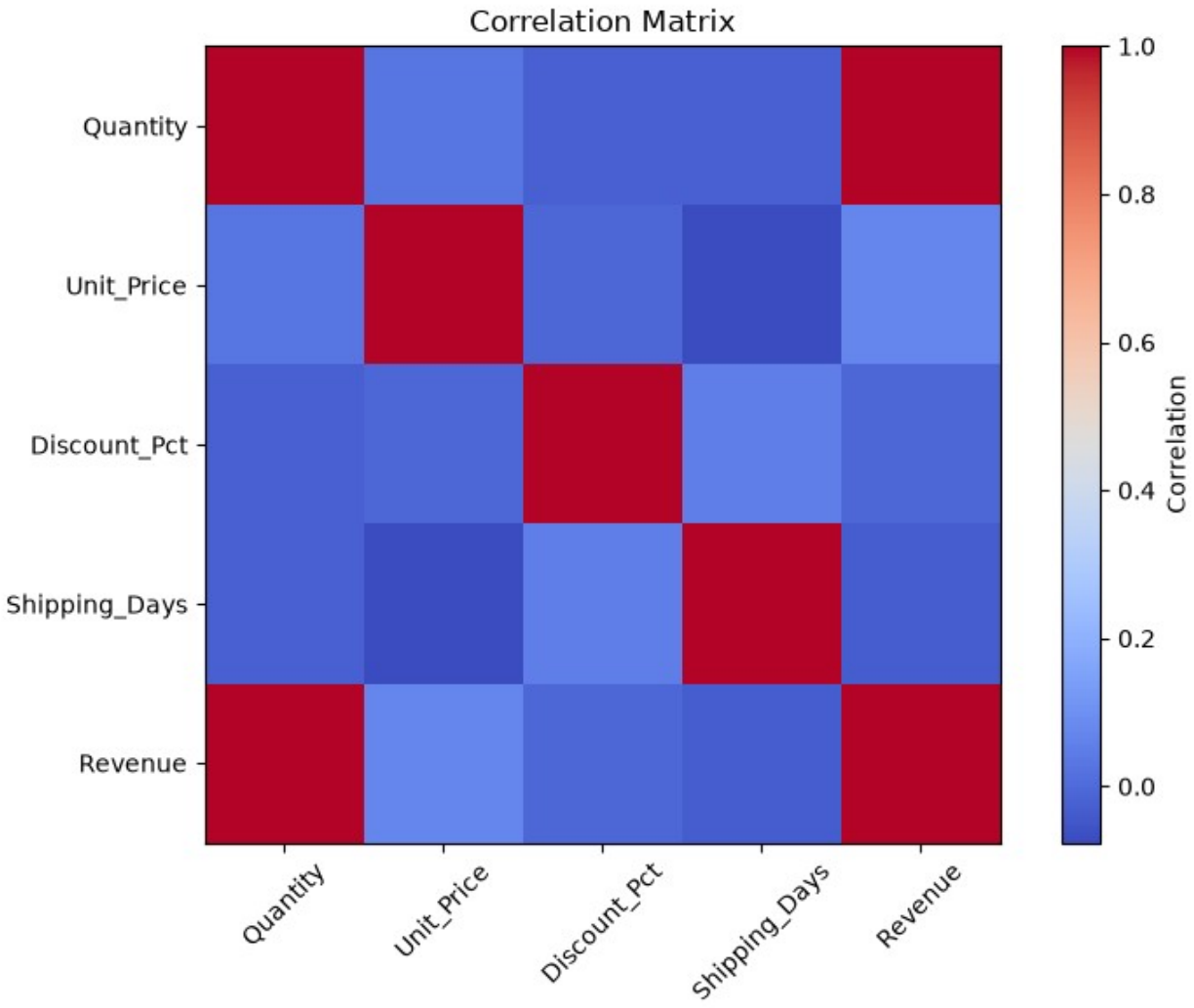
plt.colorbar(label='Correlation')

plt.xticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns,
    rotation=45
)

plt.yticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns
)

plt.title('Correlation Matrix')

plt.tight_layout()
plt.show()
```


Correlation Matrix
Quantity
Unit_Price
Discount_Pct
Shipping_Days
Revenue
Quantity
Unit_Price
Discount_Pct
Shipping_Days
Revenue
1.0
0.8
0.6
0.4
0.2
0.0
Correlation